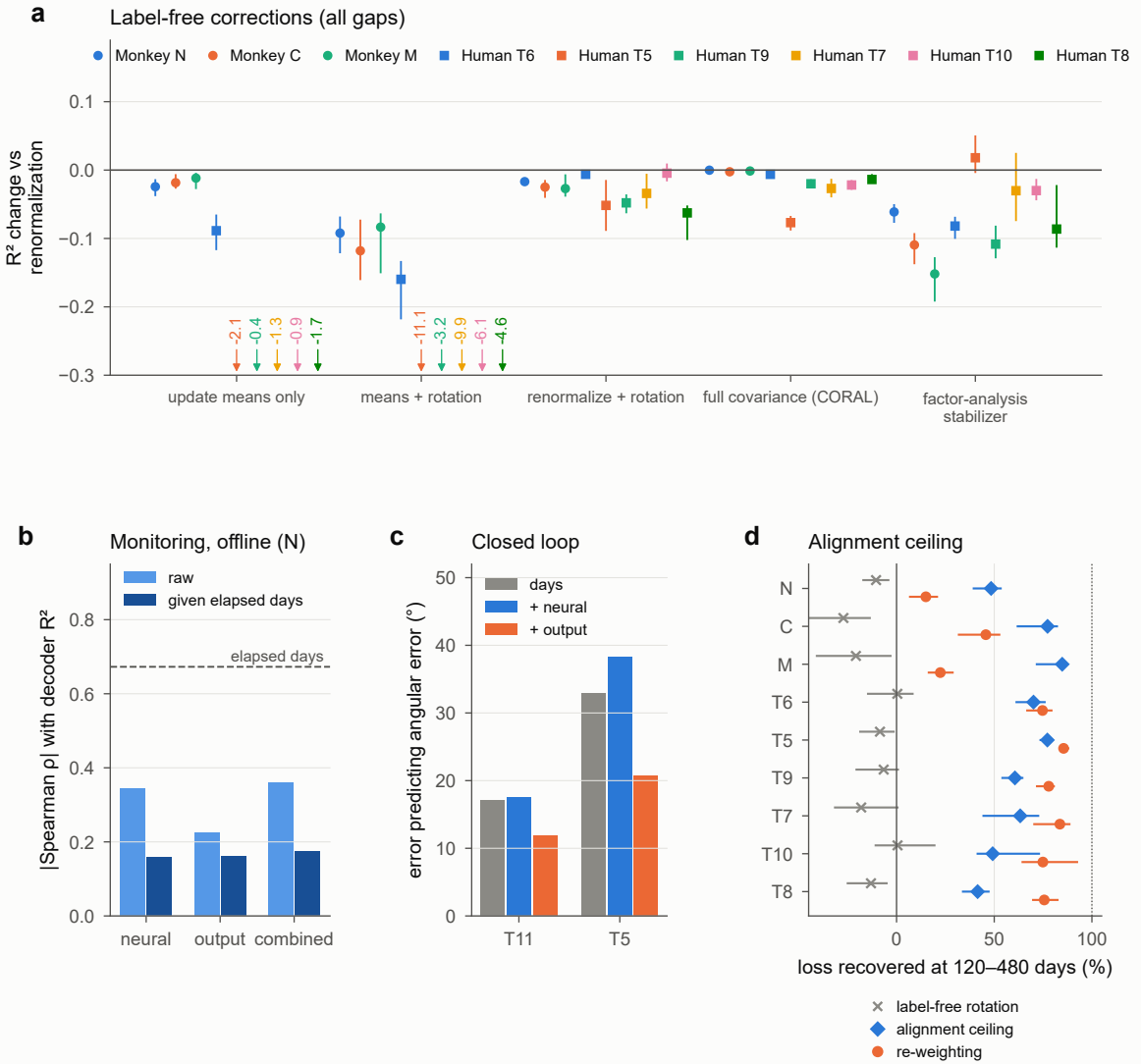

a
Label-free corrections (all gaps)
Monkey N
Monkey C
Monkey M
Human T6
Human T5
Human T9
Human T7
Human T10
Human T8
R² change vs renormalization
0.1
0.0
−0.1
−0.2
−0.3
-2.1
-0.4
-1.3
-0.9
-1.7
-11.1
-3.2
-9.9
-6.1
-4.6
update means only
means + rotation
renormalize + rotation
full covariance (CORAL)
factor-analysis stabilizer
b
Monitoring, offline (N)
raw
given elapsed days
elapsed days
|Spearman ρ| with decoder R²
0.8
0.6
0.4
0.2
0.0
neural
output
combined
c
Closed loop
days
+ neural
+ output
error predicting angular error (°)
50
40
30
20
10
0
T11
T5
d
Alignment ceiling
N
C
M
T6
T5
T9
T7
T10
T8
0
50
100
loss recovered at 120–480 days (%)
label-free rotation
alignment ceiling
re-weighting